

ActInf GuestStream 082.6 ~ Robert Worden "A Unified Theory of Language"

GuestStream | Robert Worden | 1:08:50

Hello, welcome everyone. It is October 15th, 2025 and we are in active guest stream number 82.6 continuing with the Robert Warden series. Today, Robert will be presenting and we will discuss a unified theory of language part one. So if you're watching live, please write any questions in the live chat and we'll address it after this presentation. So thank you Robert for joining to you. Okay. Uh can you hear me?

Yep.

Okay. Well, this is Robert Wen here talking about a unified theory of language. Um

this is a summation of what I've been doing in language for several years now and it's in a paper on archive which you can go to and it's quite a long paper.

It's about 50 pages actually. So this talk is in two parts

and uh I'll introduce what is coming in each part in in a minute right now.

So the talk is really about all of language and language is traditionally uh looked at in terms of these different areas. There's phonology is how sounds fit together to make words. There is syntax there is how the words fit together to make utterances uh statements and questions and so on. There is semantics how utterances express meanings.

There is pragmatics and that is how we

carry on conversations with each other.
How we understand what each other is
going on in conversation
and those are the four traditional
areas. Then of course there is learning.
How do we learn to do all of this stuff?
And finally there's evolution. How did
language evolve and why did it evolve?
And the two columns at the right speaker
and listener column are empty but they
are there to remind us that in every
question in language there are two sides
to the question. There's a speaker side
how the speakers do it and there's a
listener side how to listen understand
what's going on. And finally, um the
evolution row at the bottom. Um I
believe this uh phrase, I don't know who
said it, that nothing in biology makes
sense except in the life of evolution
and certainly language. There's a lot to
say about language evolution. And I
shall be saying that in in two parts in
both these talks.

So

here's what I'm going to cover roughly
in the first of the live streams.

language

um I shall say it works by hierarchical
basian pattern matching in other words
by minimizing free energy so it's it's a
a ripe area for the fe to be applied
it's quite a complicated area because
it's not just pattern matching many
thousands of different patterns are
involved there's we have one pattern for
every word in our native language
and the state-of-the-art now is there

are working models of phonology, syntax and semantics. The situation in pragmatics is not so clear. Um anyway, these working models have been developed in what cognitive linguistics and I shall be taking an approach that uses cognitive linguistics, not Chomsky's linguistics.

And in cognitive linguistics, every word is represented by what's called a construction. And a construction is a graph, a feature structure pattern.

And what we do when we speak language or hear it is we do Bayesian pattern matching. And the particular word in cognitive linguistics for that pattern matching is unification. And that's been developed for a long time now for a good 40 years that idea has been in development.

And we learn language by a different operation on feature structures which is called intersection. And that's a kind of complement to unification which I will talk about.

So all of that is a model of language which works at David Mars level one of requirement and level two of abstract algorithms and data structures. It's not a neural implementation and I regard the neural implementation of language as a problem yet to be solved.

So that's what I'm going to talk about in the first live stream and pragmatics I leave entirely to the second. Now just a brief summary of the second live stream of some kind of foretaste.

It's all about evolution. Language started in conversation. That was the earliest evolution of language. Simple conversations.

And language evolved by sexual selection, not just natural selection.

And that sexual selection to do a particular thing to show off our intelligence. And that's why we are so intelligent actually. And because of this language is not the neutral system that people think of. It's rather biased and it's about social status.

And in any conversation we use special patterns for mind reading for knowing what the other person thinking.

So that's theory of mind. And then I will be mainly talking about pragmatics.

Pragmatics I shall say uses the same kind of brain pattern matching as we saw in this live stream for syntax and semantics.

This model of pragmatic pragmatics addresses both the speaker side and the listener side of the problem that is new. So we learn these mind reading patterns and in those patterns we have a thing called a shadow audience which is our imagination of the other person who assesses our own status and that idea of shadow audience is the whole basis of the human sense of self.

Now the human sense of self is obviously very important because it's the basis of all human society but I shall say it's the greatest affliction. It is a great deal of trouble for us as well and I shall describe why it's trouble and

affliction and I shall finally talk a little about mindfulness which is a release from this human self. So that was a foretaste of the second live stream but enough of that we'll come back to the first now and we start with the first in evolution.

So language evolution is a question with two parts really. One part one is what is the underlying cognitive capability? What's the mechanism in the brain that does it? And second how did language actually evolve in particular one species and no other species. So this talk first part addresses only part one of the question and the answer very simply is that language is hierarchical basian pattern matching that as you know is a capability that exists in all mammals.

So hierarchical basin pattern matching in animals is the set is this sort of hierarchy bottom level we have features of sense data edges sounds and motion and so on. Next level up, those features are integrated into parts of objects, things like limbs and leaves.

Next level up, the brain classifies these into whole objects like food or animals or plants. And finally, at the very top level, we have the full generative model of three-dimensional space around the animal which is the basis of consciousness.

So that is what the hierarchy looks like in typical applications of the free energy principle on animal cognition and

there are many FE models of this and as you know it's a joint top down and bottom up inference process uh getting the two consistent with each other and there are FE neural implementations of that. So a lot of work in the FEP on that. Now going on to language.

This is how language has a similar hierarchy. The bottom level hierarchy, parts of words, syllables if you like.

Next level up, those integrate into words.

Next level up, those words are integrated into utterances. And finally, individual utterances are pulled together in conversations.

So that's the hierarchy for language and joining the levels of the hierarchy of the different topics in language.

Phenology puts phone names together and make words. Syntax and semantics put words together to make utterances. And finally, pragmatics is how utterances are put together to make conversations.

Now,

as I say, I think this is a very fertile area for the FE.

Are there any FE models of this? Now, as far as I know, there aren't because most of the work on the in the FE on language has been in the Chskin tradition rather than in a cogen linguistics tradition, which is much more akin to this hierarchical pattern matching idea.

It's quite hard to do. There are lots of patterns to be matched, but basically this is an FE application and I don't think it's been done yet in cognitive

linguistics.

There have been FE papers on chumpskin

linguistics but um

those are different. Now there is a

longrunning dispute or feud between

chskin linguistics and so-called

cognitive linguistics. Now cognitive

linguistics on the left has been

developed since about 1975 or 80 by a

large number of people. Gerald Gazdar,

Kaplan, Ronald Langaka

and many others. And it uses

unificationbased grammarss. It makes

computational models. It has

constructions which are called fe which

are feature structures.

And each construction has two poles

which I shall go through all of this in

in some detail and that is cognitive

linguistics. On the other side there is

generative jammer grammar and there is

been a long running dispute really

between workers in cogn linguistics and

followers of chsky and generative

grammar. You will recognize many of the

uh memorable slogans of generous

generative grammar. It started a long

time ago with transformation deep

structure.

It has this idea of poverty of the

stimulus that the learning stimulus for

a child is not sufficient to support

learning of language. So because the

stimulus is inadequate there has to be a

lot of innate grammar in the mind. This

is revealed by language universals and

there is a syntax module in the brain

which has autonomy which is a module on

its own has nothing to do with other parts of the brain.

And you know there's a chkin distinction of language competence versus language performance. And the story of language evolution is quite a strange one. The chskin story is that there was a kind of miracle mutation that language arrived on one individual called Prometheus uh in one generation about 200,000 years ago by a miracle mutation, a hugely unlikely mutation. So that if you like is chosen linguistics in a in a in a nutshell. There are there's been loads of ideas throughout the years. Now I believe that constant linguistics has been much more successful than chomskin linguistics for two reasons. It's built much more working computational models and they really do work for many different languages. and is confronted much more empirical data about language whereas chsy and linguistics they haven't built very computing very many working models and many of the core hypothesis have been shown to be incorrect in other words just go through those in a bit of a detail transformations deep transformations are no longer believed in the idea of the poverty of the stimulus is discredited there is plenty of stimulus for a child to learn language. So language universals turn out not to be universal. There are always exceptions. There is no alternative syntax. Syntax is

irrevocably mixed up with semantics if you like. There is no actual competence versus performance distinction. Performance was a kind of convenient way of being able to ignore a lot of the data if you like.

And there is no idea of a miracle mutation need to account for language.

So um I shall be firmly on the side of cognitive linguistics and I shall be advocating that work in the FE should take that approach to language.

So what is cognitive linguistics? It's also called construction grammar and this has been a successful model of language since about 1980. And in this model, every word is represented by what's called a construction.

And there are larger constructions for phrases as well as words. But each construction is at a logical level a directed acyclic graph, a DAG.

And the preconstruction has both a phonetic pole and a semantic pole. The phonetic pole consists of the sounds at least in the sounds of a word and a semantic pole which consists of the meaning of the word.

So that's what constructions are. You combine these constructions in a sentence by unifying them. So unification is the key operation and that operation handles both speech production and speech comprehension. In other words, it does both sides of the problem. The speaker's problem and the

listener's problem. And this approach handles phonology, syntax, and semantics. So, what I'm going to be talking about in this part of the live stream is really work that evolved over the past 40 years, 40 or 50 years. There's very little that's actually new here.

Um it's just

so

one of the things I shall refer to is I have made an online model of how cognitive linguistics handles syntax and semantics and the model is available here at bazelanguage.org or so I would urge you to go and look at that demonstration to see the things I'm talking about actually demonstrated in many different examples and this online model does production and understanding of sentences for a number of simple concrete sentences up to half a dozen words each. So what this model does it converts between meanings and words in both directions. Speakers go from meanings to words and listeners go from words to meaning.

And the way it does it is it does one unification for each word in the sentence. And you can see the unification happening.

It has about 100 words in it and you can use those words to make up your own sentences and see what it makes of them.

And it's also a model of learning. It can learn these words from scratch starting with no language models. So I

should be referring to this model and using diagrams from it when I talk.

Right?

So here is a construction a typical construction in construction grammarss.

It's a construction for the word girl and it's got it's a feature structure.

is a graph as you can see it's pregraph in this case and it's got various type nodes and the phonetic pole is just the sound of the word sound girl and it's actually broken up into phonemes she g uh but that's not expanded

and the semantic pole is the meaning of the word girl and the meaning of the word girl is a person one person and it's young and is female so that is a typical example of a construction in constructed grammarss.

Nouns are the simplest constructions because the phonetic pole consists of a noun consists only of the sounds. It's pure phonetics.

Now if we go to the next more complex thing for a verb like drinks here you find the phonetic pole is more complicated. The phonetic pole has the sound drinks in the middle but it also has two other things. It has a subject, the person doing the drinking, and an object, the thing being drunk. And it needs both those things to make sense.

So the venetic pole in this is more complicated. And the semantic pole is a scene in which a person is doing some drinking of a thing. And these curvy lines

are links which make the

construction into not a tree a directed
is acyclic graph. And I've pulled them
apart in order to pull apart the syntactic
pole from the semantic pole but in a
sense they share some nodes. And what
these curvy links do is they move
meaning. And when a listener is
listening to a sentence
and they unify the sound drink, the
meaning of the the meaning of the
subject must be there. The meaning of
the agent, the person doing the drinking
has to be there in advance. And it moves
that person
from there down into the final meaning
scene. Similarly, the thing being drunk
has to be there as well. The curved link
moves the meaning down into the drinking
scene. So that is the syntactic pole and
semantic pole for a more complicated
kind of word construction. In fact, all
word constructions
have this pattern uh with several links
one zero or more links across between
the phonetic pole and the semantic pole.
So how do we speak and understand
language in this model?
Um, this is the most complicated slide
in the in the talk and it illustrates
how we understand a simple sentence, a
two-word sentence. Uh, in this case,
doggy runs.
And the starting point is top middle.
And you can see the the listener
listening to this sentence, doggy runs,
starts off with only the sounds. And
there the the phone names doggy runs are
listed vertically.

And what the listener first does is it unifies this. The listener has a number of constructions in her mind and she has a construction for the word doggy. And that consists of the phone names doggy and the meaning uh which is a thing of type dog. And what the listener does is to unify this word doggy with the initial sound. And the result of that unification is here. What's happened is that the matching parts, the doggy part has matched between here and here. So they've been shaded out because they match and the meaning of the word doggy has been imported in here and the remaining parts of the sentence are unchanged. So that is a result of the first unification. And you can see what's happened. Unification is always a growing of patterns. It always makes patterns bigger.

So that's the first step of the unification. And that result of step one is pulled down here to the middle of the bottom. And the next step of unification is to take the word for runs, the construction of a run. And this has a complicated phonetic pole with both the thing running the top and then the sound runs and then a semantic pole here. And now we unify these two together and we match up on the sounds runs and runs. So what we get now is all the sounds have been matched up and are grayed out. But what we've produced the bottom here the bottom meaning scene in which it is a running act and the thing that doing doing the

running is a dog. So the listener has understood what the speaker said. Now that's a unification model of understanding.

There's a very similar unification model of language production. And in production, you've seen in understanding, you always match up the sounds. They match the phonetic poles and you add the semantic poles. In production, you go the other way around. You match up the semantic poles and you add the sound poles. And so you start with just a meaning and you end up with a whole load of sounds that you can express, you can say.

And so the order of unification has to be reversed there that you start with the verbs when you're producing and then you end up with the nouns. And again, unification is a way of growing the structure so that you grow from pure meaning to a meaning and a sound. And in this case the curvy links move the meaning upwards.

Okay. So that is the construction grammar cognitive linguistics model of how we speak sentences how we understand them and as I said you can see them in the demonstration where I've given you the link. Um finally um no I'm switch now. So how does this relate to the FEP? It relates because the operation of unification which has been around in cognitive linguistics for 40 or 50 years that is a basin pattern match. In other words, it's a minimization of free energy and I'll

explain briefly why that is.

The construction graph for a word has a certain information content and the information I is the sum of the information in all its nodes and slots.

So you can go around the graph looking at the information all the nodes and slots and you add it up and that simply is the information content but it also represents some situation in the world and the probability of that situation is 2^{-I} . Obviously the more information the more improbable the situation is.

Now when you unify two constructions A and B

that's written $C = A \cup B$. And how do you form C? To form C you as we've just seen you match up A and B wherever possible. Wherever they have the same slots and values and you're not allowed to have any conflicts when you match up.

So you keep the matching in the unification C. keep the matching parts and you keep both non-matching parts from A and B and that's why unification is a growing operation

and the result of this is that C the result C contains A and it also contains B

and C is the smallest graph which has this property.

In other words, of all the graphs which do this, C has the smallest possible information content. So it has the largest possible probability 2^{-I} . So it's the basian maximum likelihood match of a and b.

Equivalently it's the minimum free energy matching of a. So unification is just a free energy minimization process.

Right now the demonstration which I've told you about also does language learning. I'll talk briefly about how that works. There's another operation you can know on on two constructions and it's called intersection and you write the intersection by this funny upside down U $\cap$ A U B. Now, how do you form D? Again, you match up A and B wherever possible.

Now, you keep the matching part and you ignore any non-matching parts. So in other words any slots in A and B that are different you just forget those slots. So the result of that is that A the structure A contains B and the structure B contains D but D is the largest possible feature structure which has this property which is contained by both A and B. So D has the largest possible information content that's the intersection operation. Um intersection is essential for learning. The reason it's essential is that what intersection does is it projects out the common parts of two states of affairs and it throws away the rest. And that's what you need to do in learning. Uh language learning is a whole lot of individual sentences, situations you hear and you have to project out what's common between those in order to learn. And

I'll just briefly describe how that happens.

Bootstrap learning by intersection.

Suppose a young child has heard learned some words already A, B, and C, but hasn't work learned some new word X.

So she hears a sentence which contains the words A, B, and X.

That means she can do part of the understanding process unifying and B, but she can't unify X.

So she does what part of the understanding process she can do producing a large feature structure and she can also observe what's going on.

She can infer the whole meaning from the conversational context from what an adult is making a gesture or there's some food on the table or whatever it may be. She can guess the meaning on a reasonable proportion of those occasions. She gets the guess right. So she puts the partially understood sentence together with the guest meaning to produce what I call a learning example. And this is denoted by L1.

And she collects a small number of these learning examples L1, L2, L3. All with different A's and B's, but all of them having the same unknown word X in them.

Then in her mind she can do the fixture unconsciously of course she can do the intersection operation between these feature structures L1 intersect L2 intersect L3 and now what turns out and you can show this mathematically is that the intersection is just the construction for the new word. In other

words that intersection is the construction feature structure for X. So she learns a new word and and that way she can bootstrap actually from no words at all learning nouns first uh as children do to learning a full vocabulary and full speaking ability. So you can see how this works from the demonstration which here is the URL for demonstration again. So what this shows is that the chskin idea of poverty of the stimulus the idea that there's not enough information to learn from is just a myth.

Right, I think I'm getting fairly close to the end now. Um, this model of language, the cognitive linguistics model is a model at Mars level two. It's a abstract set of algorithms and data structures. The abstract data structures are the feature structures of but it's not a neural implementation.

It uses hierarchal basin pattern matching but there is no neural implementation of it yet as far as I know. Now I think one could implement this in the FP and one could start on neural implement implementation of it and I think that's a very important problem to try. I would dearly love to collaborate with everybody in doing that. I think there are a number of challenges in it and here are the challenges that there are thousands of word patterns to be matched. Not just a few buttons, a thousand. So you have to rapidly selectively retrieve a few

possible match patterns to see which one match best. You're always looking for the best fit. So there's some fast selective retrieval needed. And that if you get to the unification operation, it involves a discrete search space as well, a continuous search space. So ordinary gradient descent is probably going to be a rather inefficient way to do it. I don't know what is the efficient way to do that unification. And finally, the graphs that you use here are dynamic ones. Feature structures are dynamic tree structures, graph structures. How do you model those by static neural networks? Now, I just don't know the answer to those questions, but I would very much appreciate it if anybody wanted to collaborate on trying to produce an FE implementation of language in this way. So, I say I don't think it's been done yet because I think most of the FE work on language has been in the Chsian tradition rather than the cognitive linguistics tradition. So that is actually the end of part one which is took less time than I thought but so here's a summary of part one of this um live stream language works by hierarchical basian pattern matching and there are many thousands of patterns and in cognitive linguistics this has led to working computational models of phenology syntax and semantics and these These are in lots of different languages. It's been done in lots of different languages.

It covers up to semantic. It doesn't yet cover pragmatics. That is uh another subject for the next talk.

So

every word is a construction. It's a a feature structure graph. It has two poles, a semantic pole and a phonetic pole. The key operation is unification.

And unification is basian pattern matching or equivalently it's the minimization of free energy.

The same model can do learning by intersecting feature structures but this is all at Mars level two. The neural implementation at Mars level 3 is a problem still to be solved. I do invite anybody to collaborate on tackling that problem.

So that's what I've covered so far and pragmatics is the

key gradient which will be addressed in the next live stream. So I finished ahead of time and uh for questions I guess.

>> Thank you Robert. Do you want to share the baze language.org demo?

Um, yeah, I could I could do that. Um,

>> yeah, that might help

>> clarify.

>> Okay, let me um go to it.

Take a few minutes. Um,

>> yeah.

>> Yeah. Meanwhile, if anyone Yeah. has questions, they can write it in the chat. So I will bring that on screen.

Okay, go for it.

>> Okay, so here's the page you land on if you go to that and and slide sets and papers there. But actually to run the model, you do that and that's what the model looks like and you got various menus at the top. There's a help menu tells you how to get started and so on. And you can look at constructions for words and there's a whole load of words. So you take G right take the word like him and there you have the construction with its um phonetic pole here and as you hover over nodes you get explanation what's going on. Um, so there you can look at words. Um, and here you can look at a few sample sentences. You can make your own sentences, but I've got a few samples here. And there's the one doggy runs. There is all their fragments of sentences. Like we have a complicated one here. Girl who fell. And that's the meaning. That's girl who fell is the sound part of it and the meaning part is a scene in which a thing which is a girl does an action which is falling down. So what you can do with that sentence um if I can get it is you can try understanding that sentence. Let's see what I don't seem to have enough uh not sure. Yeah, I have to manipulate this a bit to get the buttons to show here. Right, if

I want to understand this sentence,
then we start with the whole sentence.
Girl who fell and you can either expand
or or contract some of these. This is
contracted, right? Girl who fell and
there's the word girl which you unifi
unify first. So you ask it to do a step
and it does a step of unification
and that's the result. So you've removed
or hidden the sound girl and you left
the sound who fell but you replace girl
by a scene in which there's a thing a
young female thing. Then you take the
next step
which is fell.

There's the result of your first step.
And there's the word fell. F E L for L.
And it's a person going on a downward
path. And you ask it to do this step.
And the result of that is you've
eliminated the the fell sound. You just
the only sound you got now left is who.
But here there is the girl and here is
here there is fell.

So you step again
and that is the word for who. It's a
rather complicated one. It's got a
couple of links here. It says
something who something did some action.
Then it it combines the curvy links
combine the actions.

So that's the word who and that is the
result of the second step of
unification.

Um and you ask to do the third step
and you get a final meaning which is a
person who is falling.

So that is quickly how the demonstration

does language understanding.

You can do the same thing for language generation. And finally, you can run it through an automatic learning sequence where it can start with no language, no words at all, and it builds up knowledge of first nouns and then the other parts of speech in language.

Okay.

>> How does it parse those first nouns? Or is that based on entity recognition?

>> Um, well, let me let me go to that. Um let's try and do that.

um autolearn. Uh I want to do a word sometime since I used this actually I'm we take a noun

apple

there's the construction it has to learn apple.

Now I think

we're unscripted here so may go wrong.

Um,

I can't

learn that word. That's a that's a pity.

Um,

the learn button doesn't seem to show.

Doesn't seem to be activated at the moment. So, um,

what I can do is autolearn.

So, it uses a bunch of words. There are some random learning examples.

And there are nouns.

And there are adjectives and verbs and and

there are some verbs there and some more verbs here. And um

I think

when I press this button again.

So what it does is it goes through
random learning examples.

Uh, example one is hot apple.

Example zero is Anna ran. She saw Tim
saw hot apple. My green apple fell. So
what's happening is that it's producing
the meanings for this. The child is
observing the meanings and doing this
intersection operation and the child has
there's a one example is hot apple. My
green apple fell. So the child produces
collects a number of learning examples
including the word apple and does the
intersection operation and each time the
child observes the meaning going on
which is my green apple or apple fell or
whatever it may be and observes the
sound apple. So when the child does the
intersection, it projects out the sound
apple and it projects out the meaning,
the round edible thing. And so the child
learns the word for apple. So I'm sorry
I couldn't demonstrate that in more
detail, but perhaps if you go and
experiment with the demonstration,
you'll find out um how to show more
detail there.

>> Okay. So there's a a juxtaposition
and an associative learning at the phmic
and the conceptual level and those get
unified into what you're calling a
script.

>> Yeah.

>> And the script is always the top node of
a construal of any feature structure.

>> Okay. And the scripts are also
composable

in that trains of phone names can be

recursively parsed for their for their
>> for their ling their lexical elements.
And then that gets parsed in a way that
is

>> not exactly the Chsky and sentence
diagram but more like a cognitive
ontology about the causal relationships
reflected by verbs and other parts of
speech.

Yeah, it it's built on animal cognition
where animals can recognize a scene and
recognize uh what's doing what
and that nonverbal pausing is the way
you build up the meaning part of
sentences.

Uh, if I just have a a slight um,
I saw him

for instance. So that meaning structure,
there's the sound structure and there's
the meaning structure. And the meaning
structure is a sort of pre-existing
animal cognition. Animals can pass
scenes into structures like that.

We've added the sounds.

>> What are the scenes? They they also seem
to be composable. And there's active and
inactive scenes.

>> Yeah, that's right. The the scenes with
just things in them. There are scenes
with action going on. Yeah.

>> So, the lower scene where act is false
is because it's just the fragment him.

Yeah, the lower scene is just describing
him

>> and then the the the composed scene is
saw him

>> which is an action because it has a

>> verb is action of seeing and it's time

in the in the past tense.

I mean th this particular diagram

notation

uh is in some sense not very important

because

they could you could have other ways of
passing

the world into scenes. You could have
carved the world apart in slightly
different ways. The same principles
would hold. You still get fix structures
and you still have unification as a way
of combining them.

So there's one kind of almost
introspective optimization about pattern
matching a new stimulus to previously
recognized stimulus and doing a kind of
detection whether it fits exactly to
what you've seen before or making some
kind of thresholding decision like
whether you're willing to accept a fuzzy
match or going to generate a new
template. But how do you connect that
kind of

knowledge

optimization with the reduction of
uncertainty

about the ecological semantics
like the uncertainty distribution of who
was it that saw him or what is it that
they did with him?

How do you represent those uncertainties
or questions

which the semantics of these diagrams
end up providing information on?

Well, I think the semantics of these
diagrams

is it comes from animal cognition and

animals have an ability to pass a scene into things and actions and recognize the importance of those things and actions. you know, animals sees a predator or sees something approaching. It's passed the scene into predator and and movement and so on and so forth. So all that is built on animal cognition and is not new for language. What's new for language is we pair that in arbitrary ways with sounds and we build those bigger feature structures that have a sound part as well as a meaning part.

But the whole animal thing is animal prior and animal evolution has has made animals categorize the world into those categories and we have a huge number of categories that we carve the world apart into before language.

Do you think this helps h how do you feel that this advances or helps give a new perspective or direction for linguistics which already has many ontologically structured sentence diagramming approaches and work at the the morpho syntax interface with like how phonemes get associated to to terminology.

>> Well, I I don't think this is new. In other words, all the work on linguistics, people like Jackendov who've done work on lexical semantics or many people have worked on it and there's a whole consensus of of all the

work they've done in cognitive linguistic linguistics started in the 80s and this is just pulling that together in a working computational model.

And I mean I'll tell you the the new parts of this first live stream are not the language production and language understanding that has been being going on for years in cognitive linguistics.

The new parts are the language learning part is new and the Bayesian interpretation linked to the FEP that is new.

So just summarize again how how would you connect this or numerically to the FEP? How do we go from this structural qualitative map to quantitative information dynamics?

Um well that goes back to the slide but if you build an FEV model of animal cognition

um

which is

visual cognition surveying a scene

classifying things into

features and objects

and the whole of that top- down model

and you link that to language.

The the meaning part is conventional FEP.

I get back to where

in other words um this this diagram here is

it there's loads of FE papers about this

and that is the semantics side of

language what's new is is linking it of

the word sound is the word phonetics

and that's been going on in cognitive linguistics. It's kind of not new. It's been going on for so long for since the 1980s. That's what cognitive linguistics has been done. It started before the FE and the two have converged if you like that they can be seen as basically doing the same thing in the domain of language.

Yeah, there's an initial ingress unification processing from either visual edge detection in the case of reading or phonemes in the case of listening

that move into a a possibly

>> very well behaved

composable cognitive space.

That

is what we tend to see in these hierarchical generative models like whether it's the reormalization group type models or just other other nested models or coarse grainings and

>> yeah any multiscale yeah

>> yeah multis and then part of the challenge is

um

in the situation where nested models represent

spatial or temporal enclosure,

you're dealing with a divisible

concept of space. So the the clustering from centimeter to meter is similar to the nesting of meter to kilometer.

However, so so that that tree only has like one kind of branch if this were a nested spatial model. But when we are

>> it also has the branch that the spatial model comes up to classified things like insects or animals or vegetables or whatever. And so it's that classification in into discrete things where the contact with language occurs, >> right? Language presents itself with many more kinds of edges, not just the the lumping and splitting of a of a homogeneous field. into different units of resolution. But it actually is the question of where and how the specificity and the semantics of different objects get reflected either in the biological substrate or in a computational substrate. So I'm just looking at, you know, this gray pen. It's not immediately obvious or there's multiple implementation strategies how there could be um just just to kind of give a um two options. There could be a color node where the the value of the color belief is gray. Or there could be a grayness andor there could be a greyness node where the belief quantitatively is like yes. Yeah. That that's the kind of representational ambiguity. We don't know the detail of how colors are represented in in language or in minds before language. Um but language has evolved to do the things that people find useful. In other words, if we live in a community where it's worth having shades of gray, we will develop words for shades of gray and children will learn the words for shades of gray. If

we live in a community where the only
uh contrast we need is dark gray versus
light gray, then we will develop words
for those and children will learn the
words for those and it'll be a different
structure that gets learned.

>> Yes. And then there's the sort of a um
the the natural history and the systems
engineer path whereas the natural
history can do different gain loss
change of functions to explore in that
marian or the tinberen framework.

What kinds of architectures are embodied
by biological systems? How do they
represent space? How do they represent
color? What are their constraints on
different kinds of beliefs and updating?
Whereas there's a bit different and
freer of a hand in the systems design
because even as we're just discussing
that there's there's a plurality of ways
to represent
even
basic
declarative semantics.

Yeah. And so you can have

>> Yeah. There's a whole story here that I
haven't gone into at all. That is that
word construction
evolve over human generations by there's
a natural selection process for words.
In other words, if a word is making a
useful distinction, if the word pen is
useful in our society, then that
construction survives as it replicates
through generations.

If the word quill is no longer useful,
then it goes extinct. So that natural

selection process is superimposed on real natural selection and happens much faster. And that's the way languages change that the constructions evolve by a very fast selection process.

>> Yeah. to to that and I think on the learning and evolutionary so within and across generational constraints

there's an analog of drift selection mutation balance where you could have one linguistic environment where there's a flat distribution you have drifts and you just have terms just change, nouns change, and things swap out.

And then the other extreme would be a heavily selected linguistic environment where there's more of a um

like truncating selection on incorrect language.

>> I I believe we're more at the heavily select for linguistics. We're more at the heavily selective end. In other words, there are strong selection pressures on words and all sorts of selection pressures. Useful meanings, simple pausing, lack of ambiguity and so on. So words come and word word species come into being and get extinct very rapidly within 10 generations or

>> Yeah. Uh uh agreed. and also that even the generational change we observe is built on a embodied causal substrate that's older than the mamleon. Um I'm really curious about what you said in

the beginning about the shadow audience
as our human sense of self. So how would
you relate that to theory of mind
thinking through other minds and what
does that mean that there's a shadow
audience?

This is the part that interests me most,
I guess, and it is for the next live
stream, but I'll give you a flavor of
it. When we're talking to someone, any
conversation,
we have to reply appropriately to what
the conversational partner is saying. We
have to work out what their what what
their intention is. So, when I ask you a
question,

I am assuming that you will know that I
need to know the answer.

So that's the theory of mine. I I ask a
question, I immediately infer that, you
know, I want an answer. And that's a
typical piece of theory of mine. But
because language is about social status,
I'm thinking not only about what do you
think about this conversation? I'm
thinking, what do you think about my
social status? Do you think I'm being
impolite? Do you think I'm being
insincere

or whatever? And so that's what goes on
in conversation with you. You are the
real audience. I use my theory of mine
which is basian pattern matching to
infer what I think you are thinking
right now. So that's when I'm speaking
to you live. When I'm offline thinking
myself, I'm imagining what what am I
going to say to Dan this afternoon?

Then I imagine you and I imagine what I say and I imagine your reaction to it. Dan will be offended. Dan will think I'm stupid or something like that. So there you are my shadow audience in my mind. I've conjured up a shadow audience and I think what's that shadow audience going to think of me? So that's the origin of my sense of myself is what I think other people are going to think about me.

So that shadow audience was born in conversation. You know it evolved in conversation because I have to carry on conversations with people. I need to rehearse those conversation in private thought. So when I rehearse any conversation I have a shadow audience and I imagine what are people going to think of me if I say that.

So that's all for what the next talk is about.

>> Yeah, that's quite a line

like how will I be seen or understood but then who am I?

>> Yeah. And I believe the human sense of self which is much more developed than chimpan chimpanzees have a certain sense of self. They can see themselves in a mirror. But monkeys can't even recognize themselves in a mirror. I think most animals just don't have a self in that human sense. And the human sense of self was born with language.

And it keeps society functioning together. I mean, that's why society works because we're all thinking about what other people are going to think of

us.

And that doesn't necessarily need to entail an individual level emulation like explicit counterfactual theory of mind. It can also get

I would I don't know about sublimated or sedimented into just heuristics for how we engage with groups.

>> It it I think it's not so much heuristic.

It's always quick subconscious pattern matching. If I say that, he will think

this and it's not a thought I have. It's something that happens before I say it and stops me saying it. So, it's very

quick and it's learned and it's

sometimes wrong. You know, I think

people who admire will I think if I buy a Ferrari, people will will admire me.

Now that thought is actually wrong, but it leads a lot of people to buy Ferraris.

Yeah. And from this belief updated hierarchical generative model

it is possible to

you can imagine a variety of other mental operations

like you can sample phone names or sentences from lower down in the tree that are consistent with the upper part.

You could imagine varieties of counterfactuals

of of how things could be differently.

You could combine this with a action model with a with a causal world model that could connect to those counterfactuals.

And then we would basically be in the setting of calculating the expected free

energy and assessing different courses of action in terms of their epistemic and pragmatic value.

Yeah, I mean the these things all filtering up and down the hierarchy basically from the phonemes of what somebody says to you to the top of your hierarchy of the self-esteem my position in society and they're all linked in one big preconscious inference.

This pattern matching is going on the whole time and it's subconscious basically. Well, you have consciousness here at the 3D generative model layer, but

>> yeah,

>> of course,

>> there's a lot of consciousness around, but the actual understanding a sentence, you're doing unifications about five per second, and that's too fast for conscious thought.

And so, you're also having ego reactions at at two or three per second. You're having eager reactions faster than conscious thought.

It's all quick basian pattern matching big picture. I yeah I am still left with the that pattern matching is just this internal kind of library science almost like there's a request coming in and there's going to be some internal process to match and either accept a stringent or a fuzzy match or make a new entry.

However, there's a second level semantics which is the reduction of

uncertainty, minimization of surprise
regarding

the message of that request.

Yeah. Does does that message fit into
the context right now? Um that's the top
down fitting
top down and bottom bottom ups happening
all the time.

So what will we look forward to in two
weeks?

Um basically

it's how language evolved, why language
evolved

and language as I say language evolved
by sexual selection to show off
intelligence and get mates. That's
basically it. So because it's showing
off intelligence, it's all about social
status in the group and the theory of
mind part of language carrying on
conversation all about self-esteem.

Um

and basically pragmatics

how we carry on conversations has been
if you like the poor relative of
linguistics.

Everybody sub studied syntax and
semantics and they have nice systems but
pragmatics has been very much less
developed since about 1980 and in
particular

people studying pragmatics don't look at
both sides of the problem. They haven't
looked at the speaker's problem as much
as the listener's problem. Pragmatics
has been all about what the listeners do
to understand in the context what the
person is saying and it's not been about

what the speaker does to get himself understood in the conversation. So the next lecture is largely about pragmatics and evolution.

Then it leads on to the human sense of self and human nature.

>> Very cool. Anything else you want to say or or ask on this one?

I think it's an awesome unification work and I I out of the corner of my eye see the kinds of graph technology developments that would help us make these types of agentic or orchestrated knowledge graphs that have elements of linguistic semantics and quantitative graphical models.

and those methods

um to to be able to to do assembly and parsing of those more rapidly to support some of these.

I I'll read one question from the live chat. Robert, so Mills 8102 wrote, "I'm thinking about how we have different modes for different environments, home, workplace, recreation, etc. I wonder what the effects are of maintaining more or less of these modes on someone's adaptability.

I in in the context of language I think that's absolutely right. We do have different personalities and different ways of presenting ourselves linguistically in different environments. we have, if you like, a different vocabulary for work versus home versus sport or whatever it may be. And so that

is one dimension uh that when we go to a different environment, we switch in a different set of prior about what words we're likely to hear and what people are likely to mean by

Okay.

Well, people

>> can I just end with one appeal? I think lang end with an appeal that language I think is a hugely productive area for the FE and I would very much like to be involved in any FE implementation of language along the lines I've sketched and to try and take it right down to the level of neural implementation that I think is a hugely promising and challenging research project and I'd welcome anybody wants to collaborate on it knows more about FE than I Awesome.

Thank you, Robert.

Till next time.

>> See you then.

>> See you.

>> Okay.